

BROHOVO, Russian FederationWMO: **259270**Lat: **59.638N**Lon: **154.303E**Elev: **5**StdP: **101.26**Time Zone: **11.00 (E11)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-33.0	-31.2	-35.8	0.1	-32.3	-34.2	0.2	-30.9	20.9	-6.4	18.3	-7.5	2.3	320	0.937

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	20.7	15.4	18.7	14.6	17.1	13.8	16.1	19.6	15.1	17.9	14.3	16.6	5.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.6	10.4	17.7	13.7	9.8	16.5	13.0	9.3	15.5	45.3	19.7	42.4	18.1	40.2	16.8	19.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	15.8	13.4	DB	-35.2	25.1	1.3	1.6	-36.1	26.2	-36.8	27.2	-37.5	28.1	-38.4	29.2	(4)
(5)			WB	-35.3	18.0	1.2	1.0	-36.2	18.7	-36.9	19.3	-37.6	19.8	-38.4	20.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.2	-18.6	-19.5	-14.8	-7.1	0.6	8.1	13.7	13.3	8.6	0.0	-9.3	-14.4	(6)	
	DBStd	12.89	7.97	7.98	7.31	5.68	2.56	3.22	2.39	1.95	2.95	3.86	5.69	6.73	(7)	
	HDD10.0	5063	887	824	767	514	290	74	1	1	59	309	579	757	(8)	
	HDD18.3	7856	1146	1058	1026	764	548	307	146	157	293	568	829	1015	(9)	
	CDD10.0	250	0	0	0	0	0	17	114	102	17	0	0	0	(10)	
	CDD18.3	2	0	0	0	0	0	0	2	1	0	0	0	0	(11)	
	CDH23.3	14	0	0	0	0	0	2	11	1	0	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	5.7	5.5	5.3	5.6	5.1	4.3	4.7	5.2	5.4	5.8	6.5	7.4	7.4	(14)	
(15) Precipitation	PrecAvg	564	35	38	31	30	26	31	53	56	61	70	65	49	(15)	
	PrecMax	865	76	108	100	87	81	107	175	134	148	158	150	183	(16)	
	PrecMin	300	2	2	3	2	1	0	1	8	2	9	4	6	(17)	
	PrecStd	135	22	28	24	22	20	25	39	32	36	39	35	35	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.3	0.3	-0.8	3.0	11.7	19.9	24.8	21.9	17.4	8.7	4.0	0.7	(19)	
		MCWB	-0.9	-0.4	-1.3	0.5	6.9	13.1	16.5	16.0	11.5	6.5	2.9	0.2	(20)	
	2%	DB	-1.7	-1.7	-2.1	1.2	8.6	16.8	21.8	19.7	15.3	7.9	2.8	-0.5	(21)	
		MCWB	-2.0	-2.2	-2.7	-0.1	5.2	11.7	16.3	15.2	11.3	5.7	2.2	-0.9	(22)	
	5%	DB	-3.3	-4.5	-3.2	0.5	6.3	14.9	19.8	18.0	13.7	6.2	0.2	-2.2	(23)	
		MCWB	-3.8	-5.0	-3.8	-0.4	3.8	10.7	15.1	14.4	10.9	4.1	-0.1	-2.8	(24)	
	10%	DB	-5.5	-7.3	-4.6	-0.1	4.7	13.2	17.8	16.6	12.5	4.9	-1.2	-4.2	(25)	
		MCWB	-6.0	-7.8	-5.1	-0.8	2.7	10.3	14.6	13.8	10.3	3.4	-1.6	-4.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	-0.3	-1.1	0.9	7.4	14.1	17.7	17.4	13.3	7.3	3.1	0.3	(27)	
		MCDB	-0.4	0.3	-0.9	1.8	11.3	18.6	22.5	20.5	15.4	8.4	3.8	0.5	(28)	
	2%	WB	-2.0	-2.1	-2.5	0.3	5.3	12.2	16.6	15.8	12.3	5.9	2.1	-0.9	(29)	
		MCDB	-1.8	-1.8	-2.2	1.0	8.2	15.4	20.4	18.7	13.7	7.5	3.0	-0.4	(30)	
	5%	WB	-3.7	-4.9	-3.6	-0.2	4.0	11.3	15.7	14.9	11.5	4.5	-0.1	-2.8	(31)	
		MCDB	-3.4	-4.6	-3.2	0.4	6.0	14.1	19.0	17.2	12.9	5.7	0.1	-2.3	(32)	
	10%	WB	-6.1	-7.8	-5.0	-0.8	2.8	10.4	14.8	14.1	10.9	3.6	-1.6	-4.6	(33)	
		MCDB	-5.6	-7.3	-4.7	-0.2	4.3	13.0	17.4	16.2	12.2	4.8	-1.2	-4.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	6.4	8.0	8.2	5.7	6.1	6.0	5.2	5.3	4.8	5.5	5.3	(35)	
		MCDBR	5.5	4.7	5.6	6.9	9.4	10.4	10.3	8.5	7.2	5.2	5.0	4.8	(36)	
	5% WB	MCWBR	5.3	4.9	5.4	6.0	6.2	5.9	5.4	5.0	4.4	4.2	5.1	4.7	(37)	
		MCWBR	5.4	4.8	5.6	5.8	8.7	8.9	8.9	6.8	5.6	4.2	5.0	4.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.238	0.277	0.332	0.407	0.437	0.407	0.455	0.423	0.346	0.323	0.273	0.235	(40)	
		taud	2.055	2.082	1.970	1.815	1.941	2.284	2.166	2.251	2.445	2.278	2.088	1.978	(41)	
	Ebn at Noon		543	716	772	769	783	825	767	764	783	660	477	379	(42)	
		Edh at Noon	46	80	123	174	166	120	132	113	79	67	45	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.40	1.25	2.83	4.48	5.22	5.43	5.17	3.81	2.60	1.29	0.44	0.19	(44)	
	RadStd		0.06	0.15	0.29	0.43	0.45	0.56	0.42	0.48	0.32	0.11	0.04	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page